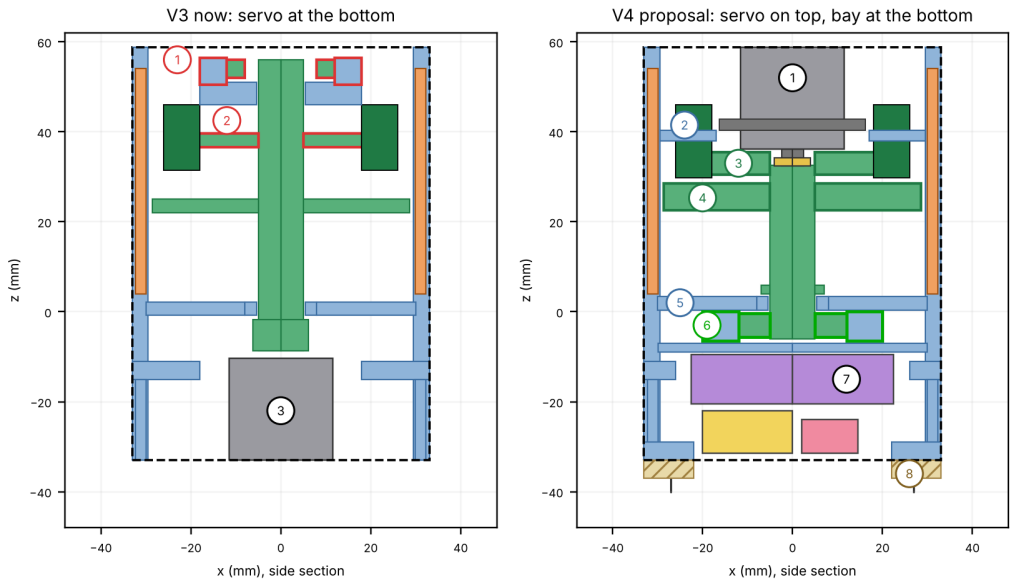
V4 concept sketch: servo on top, thicker parts, electronics bay

This is a sketch for you to inspect, not a CAD model. Nothing in Onshape has changed. Dimensions are to scale where they come from the current V3 model; the new parts are placeholders. Please read the questions on the last page before I start the CAD.

1 How I read your request

- **Thin parts:** the knob finger and the two knob stops that limit the rotor (about 2 mm wide), and the green rotor: the crown arms (3 mm thick, only 0.8 mm of material beside the screw hole) and the cam web (3 mm). All get thicker.
- **Flip:** the servo goes on **top** of the airbrake (output shaft pointing down). The rotation limiter (finger + stops) therefore moves to the **bottom** of the rotor, flipped.
- **Bottom end:** the module screws onto the top of the transition on the BT80 side, with vertical screws.
- **New bay** at the bottom for the altimeter, a small battery for the servo, and a regulator giving the servo the right voltage and current.
- **Same size:** 91.8 mm tall, 66 mm OD (the dashed box). Fins, cams, latch and tube slits stay as they are.

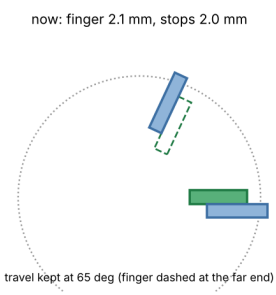
Same overall size: 91.8 mm tall, 66 mm OD (dashed box), fins unchanged (orange). SKETCH, not CAD

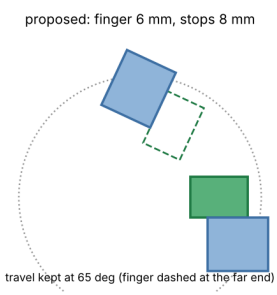


Side sections to scale. Left: V3 now. Right: V4 proposal.

#	V3 now (left)	#	V4 proposal (right)
1	knob finger and stops: about 2 mm wide	1	SG90 flipped on top, its top flush with the housing top (z 58.8), shaft pointing down
2	crown arms: 3 mm thick, 0.8 mm walls at the screw	2	ledges from the support strips carry the servo flange (M2 screws)
3	SG90 at the bottom, flush with the housing bottom	3	crown arms 5 mm thick with a 10 mm boss round the screw, now just below the servo
		4	cam web 6 mm thick (the fin follower is 9 mm tall)
		5	lower bearing (3 mm) with a collar above it: the rotor now hangs from the servo horn, the collar stops it dropping
		6	limiter moved to the bottom, flipped and thick: finger 6 mm, stops 8 mm, full-height webs
		7	bay ceiling and two electronics layers (altimeter above, battery + regulator below); base ring opened to R26
		8	bottom flange, 4 M3 up through the transition on the BT80 side

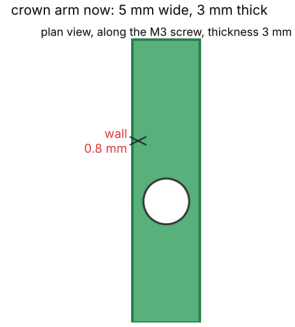
2 The thicker parts

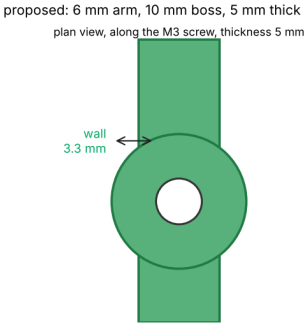




Why they snap

- finger and stops are about 2 mm wide in the printed layer direction: a single impact at the end of travel loads them like a small cantilever
- the crown arm is 5 mm wide and 3 mm thick with a 3.4 mm screw hole: only 0.8 mm of material each side
- the cam web is 3 mm thick





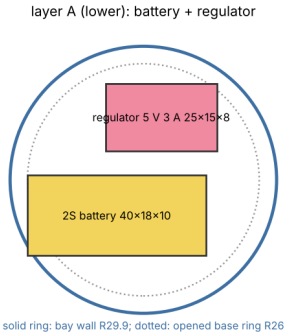
What gets thicker

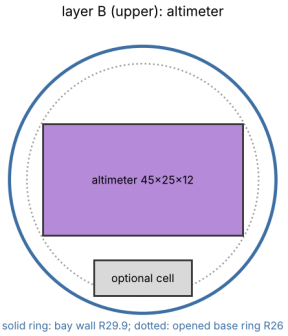
- limiter finger: 2 → 6 mm wide, 4 → 5 mm tall
- limiter stops: 2 → 8 mm wide, 6 → 9 mm tall, with a full-height web into the housing
- crown arms: 3 → 5 mm thick, 10 mm boss round the screw, fillets into the shaft
- cam web: 3 → 6 mm (follower is 9 mm)
- (printed so the layers run along the arm)

Plan views. The 65° travel is kept: the finger touches one stop at rest and the other at the end of travel.

3 The electronics bay

Electronics bay (plan views from below). Sizes are placeholders: tell me your real parts





Will it fit?

All parts side by side use about 90 % of the bay floor: impossible in one layer, so two layers.

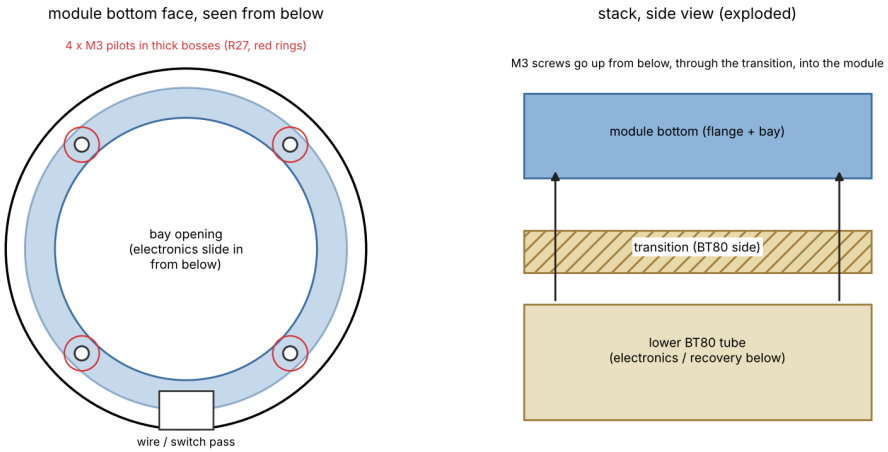
Height needed (typical parts): 10 (battery) + 12 (altimeter) + 3 (trays and gaps) = 25 mm

Available at the same overall length: z -33 to -9 = 24 mm (base ring opened up to R26).

So it is borderline: it depends on YOUR altimeter and battery. Option B below if it does not fit.

Plan views from below. All sizes are placeholders for typical parts.

4 The screwed transition



The module bottom face with four vertical M3 bosses, and the stack.

5 Two options for the bay height

	Option A: keep 91.8 mm (my default)	Option B: allow a little more
Bay height	z –33 to –9 = 24 mm, base ring opened to R26	about 24 mm + whatever your parts need, in 1 mm steps
Fits	typical parts need about 25 mm (battery 10 + altimeter 12 + trays 3): **borderline**	comfortable
Overall	91.8 mm × 66 mm, unchanged	about 92–96 mm × 66 mm
Depends on	your real altimeter and battery sizes	same

6 Power for the servo (suggestion)

- SG90 (4.8–6 V): about 0.1–0.25 A moving, about 0.7 A at stall. A 5 V regulator rated 3 A (a small step-down “UBEC” module) gives about four times the stall current.
- Battery: a 2S LiPo (7.4 V, roughly 300–450 mAh) feeding that regulator is the usual choice; a 1S cell would need a boost converter. A capacitor (about 470–1000 µF) at the servo supply and a power switch or arming jumper reachable through the transition are cheap additions.
- At 5 V the SG90 stall torque is 0.176 N·m (0.216 N·m at 6 V); the torque check in the V3 guide assumed 5 V.
- Most altimeters have no servo output. If yours does not, a small microcontroller board has to sit between the altimeter and the servo: it takes bay space too (about 20 × 10 mm).

7 Risks I can already see

- With the servo flipped, the rotor **hangs from the horn** and the top bearing is gone; the lower bearing, the collar and the limiter have to keep it in place and centred. I will check this in CAD.
- The catch caps move down about 6 mm at the arm end and become taller clips; their screw boss has to move inboard to clear the fin follower. The latch and cam timing must be re-measured afterwards.
- The servo cable exits at the top of the case and has to run down a channel inside the support strip to the bay.
- The bay is only about 1.4 mm of wall away from the outer skin all round; very little room for anything I have not listed.

8 Questions before I build it

- **Which parts are “the part at the top that restricts rotation”?** I assumed the knob finger and the two stops. If you meant something else, say so.
- **What is “the transition on the BT80 side”?** A printed coupler ring that you already have (dimensions?), or something I should design? I assumed a flat ring with four M3 holes, bonded in the BT80 tube, with the module screwed down onto it.

- **Which altimeter** (make, size) and **how should it command the servo**? Does it need its own small battery?
- **Which battery** (type, mAh, size)? If it is bigger than 40 × 18 × 10 mm, Option B is probably needed.
- **Is the servo end (top) the nose side?** I assumed yes.
- **Option A or B** for the bay height?

If you do not answer, I will use the assumptions above, Option A, and typical part sizes, and tell you exactly where they do not fit.